

Lab 6: Liquid Crystal Displays (LCD) with Arduino

1. Collect components (LCD Screen, potentiometer, LM35 temperature sensor)
2. Connect components according to the following figure
3. Writing program in Arduino IDE

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4. #include <LiquidCrystal.h>
5. int sensorpin = A0; // Declare A0 pin as sensorpin
6. int temperature=0; // Declare a variable named temperature
7. LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
8. void setup() {
9.   pinMode(sensorpin, INPUT); // Declare sensorpin as input pin
10.  lcd.begin(16, 2); // set up the LCD's number of columns and rows:
11.  lcd.print("Temperature:"); // Print a message to the LCD.
12.}
13. void loop() {
14.  temperature=analogRead(sensorpin); // Read the analog value of sensorpin and keep
    the value inside a variable named temperature
15.  temperature = temperature * 0.48828125; // 5/1023*100, Resolution of ADC=5/1023;
    Change of Vout of LM35 per degree centigrade=0.1mV
16.  lcd.setCursor(0,1);
17.  lcd.print(temperature); // Show the temperature in LCD
18.  lcd.print((char)223); // command for printing ° sign in LCD
    Ref: http://forum.arduino.cc/index.php?topic=19002.0
19.  lcd.print("C"); // C for Celcius
20.}
```



